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Isolation structures in semiconductor devices

Abstract

A semiconductor device with isolation structures and a method of fabricating the same are disclosed. The semiconductor device includes first and second FETs, an isolation structure, and a conductive structure. The first FET includes a first fin structure, a first array of gate structures disposed on the first fin structure, and a first array of S/D regions disposed on the first fin structure. The second FET includes a second fin structure, a second array of gate structures disposed on the second fin structure, and a second array of S/D regions disposed on the second fin structure. The isolation structure includes a fill portion and a liner portion disposed between the first and second FETs and in physical contact with the first and second arrays of gate structures. The conductive structure is disposed in the liner portion and conductively coupled to a S/D region of the second array of S/D regions.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Patent Application No. 63/322,532, titled “Semiconductor Device with Vertical Local Interconnect and Method for Forming the Same,” filed on Mar. 22, 2022, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

(1) With advances in semiconductor technology, there has been increasing demand for higher storage capacity, faster processing systems, higher performance, and lower costs. To meet these demands, the semiconductor industry continues to scale down the dimensions of semiconductor devices, such as metal oxide semiconductor field effect transistors (MOSFETs), including planar MOSFETs, fin field effect transistors (finFETs), and gate-all-around FETs (GAA FETs). Such scaling down has increased the complexity of semiconductor manufacturing processes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of this disclosure are best understood from the following detailed description when read with the accompanying figures.

(2) FIGS. 1A-1H illustrate isometric, cross-sectional, and top-down views of a semiconductor device with isolation structures, in accordance with some embodiments.

(3) FIGS. 2-5 illustrate top-down views of other semiconductor devices with isolation structures, in accordance with some embodiments.

(4) FIG. 6 is a flow diagram of a method for fabricating a semiconductor device with isolation structures, in accordance with some embodiments.

(5) FIGS. 7, 8A-15B, and 16A-22F illustrate isometric and cross-sectional views of a semiconductor device with isolation structures at various stages of its fabrication process, in accordance with some embodiments.

(6) Illustrative embodiments will now be described with reference to the accompanying drawings. In the drawings, like reference numerals generally indicate identical, functionally similar, and/or structurally similar elements. The discussion of elements with the same annotations applies to each other, unless mentioned otherwise.

DETAILED DESCRIPTION

(7) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the process for forming a first feature over a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. As used herein, the formation of a first feature on a second feature means the first feature is formed in direct contact with the second feature. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(8) Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(9) It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “exemplary,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.

(10) It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by those skilled in relevant art(s) in light of the teachings herein.

(11) In some embodiments, the terms “about” and “substantially” can indicate a value of a given quantity that varies within 5% of the value (e.g., $\pm 1\%$, $\pm 2\%$, $\pm 3\%$, $\pm 4\%$, $\pm 5\%$ of the value). These values are merely examples and are not intended to be limiting. The terms “about” and “substantially” can refer to a percentage of the values as interpreted by those skilled in relevant art(s) in light of the teachings herein.

(12) The fin structures disclosed herein may be patterned by any suitable method. For example, the fin structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Double-patterning or multi-patterning processes can

combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fin structures.

(13) The present disclosure provides example semiconductor devices with stacked FETs (e.g., stacked gate-all-around (GAA) FETs) and vertical interconnect structures between adjacent stacked FETs. The present disclosure further provides example methods of forming the semiconductor devices. With the use of stacked FETs, the device density of ICs can be increased without aggressively scaling down the devices and compromising the electrical isolation between the devices in the IC. In some embodiments, each of the stacked FETs can include a stack of different conductivity type GAA FETs and/or can include a stack of the same conductivity type GAA FETs. Each of the stacked FETs can further include a channel isolation layer. The channel isolation layer can electrically isolate the channel regions of the stacked FETs from each other.

(14) In some embodiments, the vertical interconnect structures (also referred to as “conductive bridge structures” and “through-via structures”) can provide electrical connections between top and bottom GAA FETs in a stacked FET. In some embodiments, the vertical interconnect structures can provide electrical connections between top GAA FETs and an interconnect structure on a back-side of the semiconductor device. In some embodiments, the vertical interconnect structures can provide electrical connections between bottom GAA FETs and an interconnect structure on a front-side of the semiconductor device. The vertical interconnect structures can be electrically connected to S/D contact structures of the top and/or bottom GAA FETs.

(15) The semiconductor device can further include isolation structures formed in a cut-metal-gate (CMG) process to “cut” long metal gate structures, extending over two or more of the stacked FETs, into shorter gate portions and to electrically isolate adjacent stacked FETs from each other. One or more of the isolation structures can include dielectric fill portions and dielectric liner portions. The vertical interconnect structures can be formed in the dielectric liner portions to reduce the device area occupied by the isolation structures and the vertical interconnect structures in the semiconductor device compared to the device area occupied by vertical interconnect structures formed adjacent to isolation structures. The formation of the vertical interconnect structures in the isolation structures also relaxes the dimensional constraints on the vertical interconnect structures. Larger vertical interconnect structures can be formed in the isolation structures, which reduces the resistance of the vertical interconnect structures by about 60% to about 90% compared to the resistance of vertical interconnect structures formed adjacent to isolation structures.

(16) Furthermore, the vertical interconnect structures can be formed in the isolation structures with fewer process steps than the number of process steps involved in forming vertical interconnect structures adjacent to isolation structures. For example, the isolation trenches (also referred to as “metal cuts”) formed using a photolithographic process and an etch process during the CMG process can be used to form both the isolation structures and the vertical interconnect structures. Portions of the isolation trenches can be filled with a dielectric material to form the dielectric fill portions. Other portions of the isolation trenches can be lined with the dielectric material to form the dielectric liner portions, which can be subsequently filled with a conductive material to form the vertical interconnect structures. Thus, a single photolithographic process and a single etch process can be used to form both the vertical interconnect structures and the isolation structures, instead of the multiple photolithographic processes and multiple etch processes used for forming vertical interconnect structures adjacent to isolation structures.

(17) FIG. 1A illustrates an isometric view of a semiconductor device **100**, according to some embodiments. FIGS. 1B, 1C, 1D, 1E, 1F, and 1G illustrate cross-sectional views of semiconductor device **100** along lines A-A, B—B, C-C, D-D, E-E, and F—F, respectively, of FIGS. 1A-1H,

according to some embodiments. FIG. 1H illustrates a top-down view of semiconductor device **100**, according to some embodiments. FIGS. 1B-1H illustrate views of semiconductor device **100** with additional structures that are not shown in FIG. 1A for simplicity. The discussion of elements with the same annotations applies to each other, unless mentioned otherwise.

(18) Referring to FIGS. 1A-1H, semiconductor device **100** can include (i) a substrate **104**, (ii) a stacked FET **102A** disposed on substrate **104**, (iii) a stacked FET **102B** disposed on substrate **104**, (iv) an isolation structure **106** disposed between stacked FETs **102A** and **102B**, and (v) a vertical interconnect structure **108** disposed in isolation structure **106**. The discussion of stacked FETs **102A** and **102B** applies to each other, unless mentioned otherwise.

(19) In some embodiments, substrate **104** can be a semiconductor material, such as silicon, germanium (Ge), silicon germanium (SiGe), a silicon-on-insulator (SOI) structure, and a combination thereof. Further, substrate **104** can be doped with p-type dopants (e.g., boron, indium, aluminum, or gallium) or n-type dopants (e.g., phosphorus or arsenic).

(20) In some embodiments, stacked FET **102A** can include (i) a fin structure **110A** disposed on substrate **104**, (ii) shallow trench isolation (STI) regions **112A** disposed on substrate **104** and adjacent to fin structure **110A**, (iii) a GAA FET **102A1** disposed on fin structure **110A** and STI regions **112A**, (iv) a GAA FET **102A2** disposed on GAA FET **102A1**, and (v) channel isolation layers **114A** disposed between GAA FETs **102A1** and **102A2**. Similarly, in some embodiments, stacked FET **102B** can include (i) a fin structure **110B** disposed on substrate **104**, (ii) STI regions **112B** disposed on substrate **104** and adjacent to fin structure **110B**, (iii) a GAA FET **102B1** disposed on fin structure **110B** and STI regions **112B**, (iv) a GAA FET **102B2** disposed on GAA FET **102B1**, and (v) channel isolation layers **114B** disposed between GAA FETs **102B1** and **102B2**. In some embodiments, fin structures **110A-110B** can include a material similar to substrate **104** and extend along an X-axis. In some embodiments, STI regions **112A-112B** can include an insulating material, such as silicon oxide, silicon nitride (SiN), silicon carbon nitride (SiCN), silicon oxycarbon nitride (SiOCN), and silicon germanium oxide.

(21) Stacked FETs **102A** and **102B** can be referred to as “complementary FETs (CFETs) **102A** and **102B**” when GAA FETs **102A1-102B1** have a conductivity type (e.g., n-type or p-type) different from that of GAA FETs **102A2-102B2**. In some embodiments, GAA FETs **102A1-102B1** can be n-type and GAA FETs **102A2-102B2** can be p-type. In some embodiments, GAA FETs **102A1-102B1** can be p-type and GAA FET **102A2-102B2** can be n-type. In some embodiments, GAA FETs **102A1-102B1** have a conductivity type different from each other and GAA FETs **102A2-102B2** have a conductivity type different from each other. In some embodiments, GAA FETs **102A1-102A2** can have the same conductivity type and GAA FETs **102B1-102B2** can have the same conductivity type.

(22) Referring to FIGS. 1A-1H, in some embodiments, GAA FET **102A1** can include (i) stacks of nanostructured layers **116A1** disposed on fin structure **110A**, (ii) gate structures **118A1-1**, **118A1-2**, and **118A1-3** surrounding nanostructured layers **116A1**, (iii) S/D regions **120A1-1**, **120A1-2**, and **120A1-3** disposed adjacent to nanostructured layers **116A1**, (iv) S/D contact structures **122A1-1** and **122A1-2** disposed on back-side of S/D regions **120A1-1** and **120A1-2**, respectively, (v) etch stop layers (ESLs) **124A1** disposed on S/D regions **120A1-1**, **120A1-2**, and **120A1-3**, (vi) interlayer dielectric (ILD) layers **126A1** disposed on ESLs **124A1**, and (vii) inner spacers **128** disposed adjacent to gate structures **118A1-1**, **118A1-2**, and **118A1-3**. Nanostructured layers **116A1** that are adjacent to and in direct contact with S/D regions **120A1-1**, **120A1-2**, and **120A1-3** function as channel regions. Nanostructured layers **116A1** that are disposed directly on bottom surfaces of channel isolation layers **114A**, as shown in FIGS. 1B and 1D, do not function as channel regions. Though a single row of channel regions is shown in FIG. 1B, GAA FET **102A1** can have one or more rows channel regions.

(23) In some embodiments, GAA FET **102A2** can include (i) stacks of nanostructured layers **116A2** disposed on channel isolation layers **114A**, (ii) gate structures **118A2-1**, **118A2-2**, and **118A2-3**

surrounding nanostructured layers **116A2**, (iii) S/D regions **120A2-1**, **120A2-2**, and **120A2-3** disposed adjacent to nanostructured layers **116A2** and on ILD layers **126A1**, (iv) S/D contact structures **122A2-1** and **122A2-2** disposed on front-side of S/D regions **120A2-1** and **120A2-2**, respectively, (v) ESLs **124A2** disposed on S/D regions **120A2-1**, **120A2-2**, and **120A2-3**, (vi) ILD layers **126A2** disposed on ESLs **124A2**, (vii) inner spacers **128A2** disposed adjacent to gate structures **118A2-1**, **118A2-2**, and **118A2-3**, (viii) gate spacers **130** disposed adjacent to gate structures **118A2-1**, **118A2-2**, and **118A2-3**, and (ix) gate capping layers **131A** disposed on gate structures **118A2-1**, **118A2-2**, and **118A2-3**. Nanostructured layers **116A2** that are adjacent to and in direct contact with S/D regions **120A2-1**, **120A2-2**, and **120A2-3** function as channel regions. Nanostructured layers **116A2** that are disposed directly on top surfaces of channel isolation layers **114A**, as shown in FIGS. **1B** and **1D**, do not function as channel regions. Though two rows of channel regions are shown in FIG. **1B**, GAA FET **102A2** can have one or more rows of channel regions.

(24) In some embodiments, GAA FET **102B1** can include (i) stacks of nanostructured layers **116B1** disposed on fin structure **110B**, (ii) gate structures **118B1-1**, **118B1-2**, and **118B1-3** surrounding nanostructured layers **116B1**, (iii) S/D regions **120B1-1**, **120B1-2**, and **120B1-3** disposed adjacent to nanostructured layers **116B1**, (iv) S/D contact structures **122B1-1** and **122B1-2** disposed on back-side of S/D regions **120B1-1** and **120B1-2**, respectively, (v) etch stop layers (ESLs) **124B1** disposed on S/D regions **120B1-1**, **120B1-2**, and **120B1-3**, (vi) interlayer dielectric (ILD) layers **126B1** disposed on ESLs **124B1**, and (vii) inner spacers **128** disposed adjacent to gate structures **118B1-1**, **118B1-2**, and **118B1-3**. Nanostructured layers **116B1** that are adjacent to and in direct contact with S/D regions **120B1-1**, **120B1-2**, and **120B1-3** function as channel regions. Nanostructured layers **116B1** that are disposed directly on bottom surfaces of channel isolation layers **114B**, as shown in FIG. **1D**, do not function as channel regions. GAA FET **102B1** can have one or more rows of channel regions.

(25) In some embodiments, GAA FET **102B2** can include (i) stacks of nanostructured layers **116B2** disposed on channel isolation layers **114B**, (ii) gate structures **118B2-1**, **118B2-2**, and **118B2-3** surrounding nanostructured layers **116B2**, (iii) S/D regions **120B2-1**, **120B2-2**, and **120B2-3** disposed adjacent to nanostructured layers **116B2** and on ILD layers **126B1**, (iv) S/D contact structures **122B2-1** and **122B2-2** disposed on front-side of S/D regions **120B2-1** and **120B2-2**, respectively, (v) ESLs **124B2** disposed on S/D regions **120B2-1**, **120B2-2**, and **120B2-3**, (vi) ILD layers **126B2** disposed on ESLs **124B2**, (vii) inner spacers **128B2** disposed adjacent to gate structures **118B2-1**, **118B2-2**, and **118B2-3**, (viii) gate spacers **130** disposed adjacent to gate structures **118B2-1**, **118B2-2**, and **118B2-3**, and (ix) gate capping layers **131B** disposed on gate structures **118B2-1**, **118B2-2**, and **118B2-3**. Nanostructured layers **116B2** that are adjacent to and in direct contact with S/D regions **120B2-1**, **120B2-2**, and **120B2-3** function as channel regions. Nanostructured layers **116B2** that are disposed directly on top surfaces of channel isolation layers **114B**, as shown in FIG. **1D**, do not function as channel regions. GAA FET **102B2** can have one or more rows of channel regions.

(26) In some embodiments, the gate structures of stacked FET **102A** and **102B** can have substantially equal gate lengths along an X-axis. Gate structures **118A1-1**, **118A1-2**, **118A1-3**, **118B1-1**, **118B1-2**, and **118B1-3** can be referred to as “bottom gate structures” and gate structures **118A2-1**, **118A2-2**, **118A2-3**, **118B2-1**, **118B2-2**, and **118B2-3** can be referred to as “top gate structures.”

(27) The discussion of the elements of GAA FET **102A1** applies to the corresponding elements of GAA FET **102B1** and the discussion of the elements of GAA FET **102A2** applies to the corresponding elements of GAA FET **102B2**, unless mentioned otherwise. As used herein, the term “nanostructured” defines a structure, layer, and/or region as having a horizontal dimension (e.g., along an X- and/or Y-axis) and/or a vertical dimension (e.g., along a Z-axis) less than about 100 nm, for example about 90 nm, about 50 nm, about 10 nm, or other values less than about 100 nm.

In some embodiments, nanostructured layers **116A1**, **116A2**, **116B1**, and/or **116B2** can have be in the form of nanosheets, nanowires, nanorods, nanotubes, or other suitable nanostructured shapes. (28) In some embodiments, ESLs **124A1-124A2**, ILD layers **126A1-126A2**, inner spacers **128A1-128A2**, and gate spacers **130** can include an insulating material, such as silicon oxide, silicon nitride (SiN), silicon carbon nitride (SiCN), silicon oxycarbon nitride (SiOCN), and silicon germanium oxide.

(29) Channel isolation layers **114A** can electrically isolate channel regions of GAA FET **102A1** from the overlying channel regions of GAA FET **102A2**. In some embodiments, channel isolation layers can include a dielectric material with a dielectric constant ranging from about 3 to about 25. In some embodiments, the dielectric material can include SiO₂, SiN, silicon oxynitride (SiO_xN_y), silicon oxycarbon nitride (SiO_xCyN_z), HfO₂, ZrO₂, or a combination thereof. In some embodiments, the dielectric material can include a material with a dielectric constant lower than the dielectric constant of SiO₂ (about 3.9), such as hydrogenated carbon-doped silicon oxide (SiCOH) (dielectric constant ranging from about 2.7 to about 3.3), fluorosilicate glass (FSG) (dielectric constant of about 3.5 to about 3.9), nanopore carbon doped oxide (CDO), black diamond (BD), a benzocyclobutene (BCB) based polymer, an aromatic (hydrocarbon) thermosetting polymer (ATP), hydrogen silsesquioxane (HSQ), methyl silsesquioxane (MSQ), poly-arylene ethers (PAE), diamond-like carbon (DLC) doped with nitrogen, and combinations thereof. In some embodiments, thickness of channel isolation layers **114A** can range from about 10 nm to about 30 nm for adequate electrical isolation between channel regions of GAA FETs **102A1** and **102A2** without compromising device size and manufacturing cost. In some embodiments, channel isolation layers **114A** can have side surfaces with linear side profiles (shown in FIGS. **1B**, **1D**, and **1F**), faceted side profiles (not shown), or tapered side profiles (not shown).

(30) In some embodiments, nanostructured layers **116A1** and **116A2** can include semiconductor materials similar to or different from substrate **104**. In some embodiments, nanostructured layers **116A1** and **116A2** can include Si, SiAs, silicon phosphide (SiP), SiC, SiCP, SiGe, Silicon Germanium Boron (SiGeB), Germanium Boron (GeB), Silicon-Germanium-Tin-Boron (SiGeSnB), a III-V semiconductor compound, or other suitable semiconductor materials. Though rectangular cross-sections of nanostructured layers **116A1** and **116A2** are shown, nanostructured layers **116A1** and **116A2** can have cross-sections of other geometric shapes (e.g., circular, elliptical, triangular, or polygonal).

(31) The discussion of gate structure **118A1-1** applies to gate structures **118A1-2** and **118A1-3**, and the discussion of gate structure **118A2-1** applies to gate structures **118A2-2** and **118A2-3**, unless mentioned otherwise. In some embodiments, gate structures **118A1-1** and **118A2-1** can include (i) interfacial oxide (IL) layers **132** disposed on nanostructured layers **116A1** and **116A2**, and (ii) high-k (HK) gate dielectric layers **134** disposed on IL layers **132**. In some embodiments, IL layers **132** can include silicon oxide (SiO₂), silicon germanium oxide (SiGeO₂), or germanium oxide (GeO₂). In some embodiments, HK gate dielectric layers **134** can include a high-k dielectric material, such as hafnium oxide (HfO₂), titanium oxide (TiO₂), hafnium zirconium oxide (HfZrO), tantalum oxide (Ta₂O₃), hafnium silicate (HfSiO₄), zirconium oxide (ZrO₂), and zirconium silicate (ZrSiO₂).

(32) In some embodiments, gate structures **118A1-1** and **118A2-1** can further include conductive layers **136A1** and **136A2**, respectively. Conductive layers **136A1-136A2** can be multi-layered structures. The different layers of conductive layers **136A1-136A2** are not shown for simplicity. Each of conductive layers **136A1-136A2** can include a WFM layer disposed on HK dielectric layer **134** and a gate metal fill layer on the WFM layer. In some embodiments, the WFM layers can include titanium aluminum (TiAl), titanium aluminum carbide (TiAlC), tantalum aluminum (TaAl), tantalum aluminum carbide (TaAlC), Al-doped Ti, Al-doped TiN, Al-doped Ta, Al-doped TaN, or other suitable Al-based materials for n-type GAA FET **102A1** or **102A2**. In some embodiments, the WFM layers can include substantially Al-free (e.g., with no Al) Ti-based or Ta-based nitrides or

alloys, such as titanium nitride (TiN), titanium silicon nitride (TiSiN), titanium gold (Ti—Au) alloy, titanium copper (Ti—Cu) alloy, tantalum nitride (TaN), tantalum silicon nitride (TaSiN), tantalum gold (Ta—Au) alloy, and tantalum copper (Ta—Cu) for p-type GAA FET **102A1** or **102A2**. The gate metal fill layers can include a suitable conductive material, such as tungsten (W), Ti, silver (Ag), ruthenium (Ru), molybdenum (Mo), copper (Cu), cobalt (Co), Al, iridium (Ir), nickel (Ni), metal alloys, and a combination thereof.

(33) In some embodiments, gate structures **118A1-1** and **118A2-1** can be electrically isolated from adjacent S/D regions **120A1-1** and **120A2-1** by inner spacers **128**. In some embodiments, gate structure **118A2-1** can be electrically isolated from adjacent S/D contact structure **122A2-1** by gate spacer **130**.

(34) In some embodiments, S/D regions **120A1-1**, **120A1-2**, and **120A1-3** can include an epitaxially-grown semiconductor material, such as Si and SiGe, and p-type dopants, such as boron and other suitable p-type dopants. In some embodiments, S/D regions **120A2-1**, **120A2-2**, and **120A2-3** can include an epitaxially-grown semiconductor material, such as Si, and n-type dopants, such as phosphorus and other suitable n-type dopants. S/D regions **120A1-1**, **120A1-2**, **120A1-3**, **120B1-1**, **120B1-2**, and **120B1-3** can be referred to as “bottom S/D regions” and S/D regions **120A2-1**, **120A2-2**, **120A2-3**, **120B2-1**, **120B2-2**, and **120B2-3** can be referred to as “top S/D regions.”

(35) Referring to FIGS. 1B, 1E, and 1G-1H, S/D contact structures **122A1-1**, **122A1-2**, **122B1-1**, and **122B1-2** can be referred to as “back-side S/D contact structures” and S/D contact structures **122A2-1**, **122A2-2**, **122B2-1**, and **122B2-2** can be referred to as “front-side S/D contact structures.” In some embodiments, each of front- and back-side S/D contact structures can include silicide layers **138**, contact plugs **140** disposed on silicide layers **138**, and nitride barrier layers **142** along sidewalls of contact plugs **140**. In some embodiments, silicide layers **138** can include titanium silicide (Ti.sub.xSi.sub.y), tantalum silicide (Ta.sub.xSi), molybdenum silicide (Mo.sub.xSi.sub.y), nickel silicide (Ni.sub.xSi.sub.y), cobalt silicide (Co.sub.xSi.sub.y), tungsten silicide (W.sub.xSi.sub.y), or a combination thereof. In some embodiments, contact plugs **140** can include conductive materials, such as cobalt (Co), tungsten (W), ruthenium (Ru), iridium (Ir), nickel (Ni), osmium (Os), rhodium (Rh), aluminum (Al), molybdenum (Mo), copper (Cu), zirconium (Zr), stannum (Sn), silver (Ag), gold (Au), zinc (Zn), cadmium (Cd), and a combination thereof.

(36) In some embodiments, the front-side S/D contact structures can be electrically connected to power supplies and/or other active devices through front-side interconnect structure (not shown) formed on GAA FETs **102A2** and **102B2**. In some embodiments, the back-side S/D contact structures can be electrically connected to power supplies and/or other active devices through back-side interconnect structure (not shown) formed on the back-side S/D contact structures. In some embodiments, S/D contact structures **122B1-2** and **122B2-2** can be electrically connected to each other through vertical interconnect structure **108**, as shown in FIG. 1G. In some embodiments, instead of both S/D contact structures **122B1-2** and **122B2-2** being connected to vertical interconnect structure **108**, S/D contact structure **122B1-2** can be electrically connected to the front-side interconnect structure through vertical interconnect structure **108** or S/D contact structure **122B2-2** can be electrically connected to the back-side interconnect structure through vertical interconnect structure **108**. In some embodiments, instead of S/D contact structures **122B1-2** and **122B2-2**, gate contact structures **144B1-2** and/or **144B2-2** can be electrically connected to vertical interconnect structure **108**, as shown in FIG. 1F. Thus, vertical interconnect structure **108** can provide flexible routing for S/D regions and gate structures in stacked FETs, such as stacked FETs **102A-102B**.

(37) Referring to FIGS. 1E and 1G, the front-side S/D contact structures can have heights H1-H4 of about 15 nm to about 35 nm and the back-side S/D contact structures can have heights H5-H8 of about 25 nm to about 35. Within these ranges of heights H1-H8, the front- and back-side S/D

contact structures can form adequate contact areas with corresponding top and bottom S/D regions without compromising the structure of the top and bottom S/D regions and the device manufacturing cost. In some embodiments, heights H1-H4 of the front-side S/D contact structures can be substantially equal to or different from each other and heights H5-H8 of the back-side S/D contact structures can be substantially equal to or different from each other.

(38) In some embodiments, the front-side S/D contact structures can have widths W1-W4 of about 5 nm to about 35 nm and the back-side S/D contact structures can have widths W5-W8 of about 25 nm to about 65 nm. Within these ranges of widths W1-W8, the front- and back-side S/D contact structures can form adequate contact areas with corresponding top and bottom S/D regions and can provide adequate landing areas for via structures (not shown) formed on the front- and back-side S/D contact structures. Furthermore, within these ranges of widths W1-W8, (i) S/D contact structures **122B1-2** and **122B2-2** can form adequate contact areas with vertical interconnect structure **108**, and (ii) S/D contact structures **122A1-1**, **122A1-2**, **122A2-1**, **122A2-2**, **122B1-1**, and **122B2-1** can form adequate contact areas with corresponding top and bottom S/D regions without overlapping with vertical interconnect structure **108** and isolation structure **106**. In some embodiments, widths W1-W4 of the front-side S/D contact structures can be formed substantially equal to each other and widths W5-W8 of the back-side S/D contact structures can be formed substantially equal to each other for the ease of fabrication. In some embodiments, widths W5-W8 of the back-side S/D contact structures can be formed greater than widths W1-W4 of the front-side S/D contact structures as fabrication of the back-side S/D contact structures are less constrained than that of the front-side S/D contact structures due to fewer elements formed on the back-side of semiconductor device **100** compared to its front-side.

(39) Referring to FIGS. **1A** and **1C-1H**, in some embodiments, isolation structure **106** can electrically isolate stacked FETs **102A** and **102B** from each other. FIG. **1H** does not show ESLs **124A2-124B2** and ILD layers **126A2-126B2** for simplicity. Isolation structure **106** can be formed in a CMG process (described in further detail below) to cut long gate structures (e.g., along a Y-axis) formed across fin structures **110A** and **110B** into shorter gate structures of stacked FETs **102A** and **102B**. In some embodiments, isolation structure **106** can include dielectric fill portions **106A** and a dielectric liner portion **106B**. In some embodiments, dielectric fill portions **106A** and dielectric liner portion **106B** can include a nitride material, such as silicon nitride.

(40) In some embodiments, dielectric liner portion **106B** can have a thickness T2 greater than thickness T1 of dielectric fill portions **106A** because of vertical interconnect structure **108** disposed in dielectric liner portion **106B**. Dielectric liner portion **106B** can surround vertical interconnect structure **108**, as shown in FIG. **1H**. The inner sidewalls of dielectric liner portion **106B** can be in physical contact with the sidewalls of vertical interconnect structure **108**, except for the sidewall portions of vertical interconnect structure **108** in physical contact with S/D contact structures **122B1-2** and/or **122B2-2** or gate structures **144B1-2** and/or **144B2-2**. In some embodiments, vertical interconnect structure **108** can include a conductive material, such as Co, W, Ru, Ir, Ni, Os, Rh, Al, Mo, Cu, Ag, Au, and a combination thereof.

(41) Vertical interconnect structure **108** can be formed in isolation structure **106** to reduce the device area occupied by isolation structure **106** and the vertical interconnect structure **108** compared to the device area occupied by vertical interconnect structures formed adjacent to isolation structures. The formation of vertical interconnect structure **108** in isolation structure **106** also relaxes the dimensional constraints on vertical interconnect structure **108**. Vertical interconnect structure **108** with greater thickness along a Y-axis can be formed in isolation structure **106** compared to vertical interconnect structures formed adjacent to isolation structures. As a result of larger vertical interconnect structure **108**, the resistance of vertical interconnect structure **108** can be reduced by about 60% to about 90% compared to the resistance of vertical interconnect structures formed adjacent to isolation structures.

(42) Furthermore, vertical interconnect structure **108** can be formed in isolation structure **106** with

fewer process steps than the number of process steps involved in forming vertical interconnect structures adjacent to isolation structures. For example, isolation trenches **1806** (shown in FIGS. **18B-18F**) formed using a photolithographic process and an etch process during the CMG process can be used to form both isolation structure **106** and vertical interconnect structure **108**, as described below with reference to FIGS. **18A-20F**. A portion of the isolation trench **1806** can be lined with a dielectric material to form dielectric liner portion **106B**, which can be subsequently filled with a conductive material to form vertical interconnect structure **108** and other portions of isolation trench **1806** can be filled with the dielectric material to form dielectric fill portions **106A**. Thus, a single photolithographic process and a single etch process can be used to form both vertical interconnect structure **108** and isolation structure **106**, instead of the multiple photolithographic processes and multiple etch processes used for forming vertical interconnect structures adjacent to isolation structures.

(43) In some embodiments, sidewalls of dielectric fill portions **106A** and dielectric liner portion **106B** along an X-axis and facing stacked FET **102A** can be substantially coplanar with each other. In some embodiments, sidewalls of dielectric fill portions **106A** and dielectric liner portion **106B** along an X-axis and facing stacked FET **102B** can be non-coplanar. In some embodiments, a portion of dielectric liner portion **106B** and a portion of vertical interconnect structure **108** facing stacked FET **102B** can be protruded from sidewalls of dielectric fill portions **106A** facing stacked FET **102B**. The protruding portions of dielectric liner portion **106B** and vertical interconnect structure **108** can be in physical contact with S/D contact structures **122B1-2** and **122B2-2**. Due to these protruding portions, S/D contact structures **122B1-2** and **122B2-2** can be electrically connected to vertical interconnect structure **108** with widths substantially equal to widths of other S/D contact structures of stacked FET **102B** that are not electrically connected to vertical interconnect structure **108**. The formation of the front-side S/D contact structures with substantially equal widths **W1-W4** and of the back-side S/D contact structures with substantially equal widths **W5-W8** reduces the complexity of fabrication, thus increasing manufacturing yield and decreasing manufacturing cost.

(44) Referring to FIGS. **1C-1H**, in some embodiments, isolation structure **106** can have a height **H9** of about 110 nm to about 170 nm to provide adequate electrical isolation between stacked FETs **102A** and **102B**. As vertical interconnect structure **108** is disposed in isolation structure **106**, and top and bottom surfaces of vertical interconnect structure **108** are substantially coplanar with those of isolation structure **106**, vertical interconnect structure **108** can have height **H9** to provide adequate electrical connection between S/D contact structures **122B1-2** and **122B2-2**. In some embodiments, dielectric fill portions **106A** can have a thickness **T1** of about 10 nm to about 20 nm to provide adequate electrical isolation between stacked FETs **102A** and **102B**.

(45) In some embodiments, dielectric liner portion **106B** can have a thickness **T2** of about 30 nm to about 40 nm and a width **W9** of about 50 nm to about 220 nm or of about 1.5 contacted poly pitch (CPP) to about 4.5 CPP. In some embodiments, 1 CPP can be about 45 nm to about 48 nm and 4.5 CPP can be greater than about 200 nm. Within these ranges of thickness **T2** and width **W9**, dielectric liner portion **106B** provide adequate electrical isolation between stacked FETs **102A** and **102B** without overlapping with their S/D regions and provide adequate space to enclose vertical interconnect structure **108**. The CPP (also referred to as “gate pitch”) is defined as a sum of a distance along an X-axis between adjacent gate structures of substantially equal gate lengths (e.g., gate structures **118B2-2** and **118B2-3**) and a gate length (e.g., GL shown in FIG. **1H**) of one of the adjacent gate structures. The CPP is also defined as a distance along an X-axis between the symmetry lines along a Y-axis of adjacent gate structures of substantially equal gate lengths.

(46) In some embodiments, dielectric liner portion **106B** can have a thickness **T3** of 5 nm to about 10 nm surrounding vertical interconnect structure **108** to provide adequate electrical isolation between vertical interconnect structure **108** and gate structures **118A1-2**, **118A1-3**, **118A2-2**, **118A2-3**, **118B1-2**, **118B1-3**, **118B2-2**, and **118B2-3**. Furthermore, in some embodiments,

sidewalls of dielectric liner portion **106B** along a Y-axis can be aligned with sidewalls of gate spacers **130** along a Y-axis to prevent misalignment of vertical interconnect structure **108** with S/D contact structures **122B1-2** and **122B2-2**. In some embodiments, standard cell boundary G-G can be aligned between sidewalls of isolation structure **106** and vertical interconnect structure **108** along an X-axis and facing stacked FET **102A**.

(47) In some embodiments, vertical interconnect structure **108** can have a thickness **T4** of about 15 nm to about 25 nm to have an aspect ratio (e.g., **H9:T4**) of about 5 to about 10 without compromising device size and manufacturing cost. Within these aspect ratio ranges, fabrication defects (e.g., voids) in vertical interconnect structure **108** can be prevented or minimized, thus improving the electrical conductivity of vertical interconnect structure **108**. Furthermore, vertical interconnect structure **108** can have a width **W10** of about 45 nm to about 200 nm or about 1 CPP (e.g., about 45 nm to about 48 nm) to about 4 CPP (e.g., greater than about 180 nm) to improve its electrical conductivity and to prevent misalignment with S/D contact structures **122B1-2** and **122B2-2**. These ranges of width **W10** can also provide adequate contact area between vertical interconnect structure **108** and S/D contact structures **122B1-2** and **122B2-2**. In some embodiments, width **W10** can be about 1 CPP to about 1.5 CPP when vertical interconnect structure **108** are electrically connected to one front- and/or one back-side S/D contact structure, such as S/D contact structures **122B1-2** and **122B2-2**.

(48) FIGS. 2-5 illustrate different top-down views of semiconductor device **100** with different configurations of isolation structures and vertical interconnect structures, according to some embodiments. FIGS. 2-5 illustrate views of semiconductor device **100** with additional structures that are not shown in FIGS. 1A-1H for simplicity. Elements in FIGS. 2-5 with the same annotations as elements in FIGS. 1A-1H are described above.

(49) Referring to FIGS. 2-5, stacked FETs **102A** and **102B** can include gate structures **118A2-4**, **118A2-5**, **118B2-4**, and **118B2-5**, S/D regions **120A2-3** and **120B2-3**, and S/D contact structures **122A2-3**, **122B2-3**, and **122B2-4** that are not shown in FIGS. 1A-1H for simplicity. The discussion of gate structures, S/D regions, and S/D contact structures of FIGS. 1A-1H applies to those of FIGS. 2-5, unless mentioned otherwise.

(50) In some embodiments, instead of isolation structure **106** and vertical interconnect structure **108**, semiconductor device **100** can have (i) an isolation structure **206** and vertical interconnect structures **208A-208B**, as shown in FIG. 2, (ii) isolation structures **306A-306B** and vertical interconnect structures **208A-208B**, as shown in FIG. 3, (iii) isolation structures **306A** and **406** and a vertical interconnect structure **208A**, as shown in FIG. 4, or (iv) an isolation structure **506** and a vertical interconnect structure **508**, as shown in FIG. 5, disposed between stacked FETs **102A** and **102B**. In some embodiments, isolation structure **106** and vertical interconnect structure **108** can be disposed between stacked FETs **102A** and **102B**, and the isolation structures and vertical interconnect structures of FIGS. 2, 3, 4, and/or 5 can be disposed in different device areas and between different pairs of stacked FETs of semiconductor device **100** similar to the pair of stacked FETs **102A-102Bs**. In some embodiments, semiconductor device **100** can have any combination of the isolation structures and vertical interconnect structures shown in FIGS. 1H and 2-5.

(51) Referring to FIG. 2, in some embodiments, isolation structure **206** can include a dielectric fill portion **206A** and dielectric liner portions **206B1-206B2**. Vertical interconnect structures **208A** and **208B** can be disposed in dielectric liner portions **206B1** and **206B2**, respectively. The discussion of dielectric liner portion **106B** and vertical interconnect structure **108** applies to dielectric liner portion **206B2** and vertical interconnect structure **208B**, respectively, unless mentioned otherwise. Isolation structure **206** can include a material similar to that of isolation structure **106** and vertical interconnect structures **208A-208B** can include a material similar to that of vertical interconnect structure **108**. Dielectric liner portion **206B1** can have thicknesses **T5** and **T6** substantially equal to thicknesses **T2** and **T3**, respectively, of dielectric liner portion **206B2**. Vertical interconnect structure **208A** can have a thickness **T7** substantially equal to thickness **T4** of vertical interconnect

structure **208B**. Isolation structure **206** and vertical interconnect structures **208A-208B** can have heights similar to height **H9** of isolation structure **106** and vertical interconnect structures **108**. Vertical interconnect structures **208A-208B** can have aspect ratios similar to that of vertical interconnect structure **108**.

(52) Vertical interconnect structure **208A** can be configured to be in physical contact with two S/D contact structures **122B2-1** and **122B2-2** and vertical interconnect structure **208B** can be configured to be in physical contact with one of S/D contact structures **122B2-1**. In some embodiments, vertical interconnect structure **208A** can have a width **W12** of about 2 CPP to about 4 CPP to improve its electrical conductivity and to prevent misalignment with S/D contact structures **122B2-1** and **122B2-2**. This range of width **W12** can also provide adequate contact area between vertical interconnect structure **208A** and S/D contact structures **122B2-1** and **122B2-2**. In some embodiments, width **W11** can be about 2 CPP to about 2.5 CPP when vertical interconnect structure **208A** are electrically connected to two S/D contact structures, such as S/D contact structures **122B2-1** and **122B2-2**.

(53) Isolation structure **206** can be configured to (i) align dielectric fill portion **206A** with S/D contact structures (e.g., S/D contact structure **122B2-3**) that are not electrically connected to any vertical interconnect structures, and (ii) align dielectric liner portions **206B1-206B2** with S/D contact structures (e.g., S/D contact structures **122B2-1**, **122B2-2**, and **122B2-4**) that are electrically connected to vertical interconnect structures (e.g., vertical interconnect structures **208A** and **208B**). In some embodiments, dielectric liner portion **206B1** can have a thickness **T5** of about 30 nm to about 40 nm and a width **W11** of about 50 nm to about 160 nm or of about 2.5 CPP to about 4.5 CPP. Within these ranges of thickness **T5** and width **W11**, dielectric liner portion **206B1** provide adequate electrical isolation between stacked FETs **102A** and **102B** without overlapping with their S/D regions and provide adequate space to enclose vertical interconnect structure **208A**.

(54) In some embodiments, thickness **T6** of dielectric liner portion **206B1** surrounding vertical interconnect structure **208A** provide adequate electrical isolation between vertical interconnect structure **208A** and gate structures **118A2-1**, **118A2-2**, **118A2-3**, **118B2-1**, **118B2-2**, and **118B2-3**. In some embodiments, sidewalls of dielectric liner portion **206B 1** along a Y-axis can be aligned with sidewalls of gate spacers **130** along a Y-axis to prevent misalignment of vertical interconnect structure **208A** with SID contact structures **122B2-1** and **122B2-2**. In some embodiments, sidewalls of dielectric liner portion **206B2** along a Y-axis can be aligned with sidewalls of gate spacers **130** along a Y-axis to prevent misalignment of vertical interconnect structure **208B** with SID contact structure **122B2-4**.

(55) Referring to FIG. 3, in some embodiments, semiconductor device **100** can have segmented isolation structures **306A** and **306B**, which are spaced apart from each other along an X-axis. In some embodiments, isolation structures **306A** and **306B** can be similar to dielectric liner portions **206B1** and **206B2**, respectively. However, unlike dielectric liner portions **206B1** and **206B2**, isolation structures **306A** and **306B** are not connected to each other through a dielectric fill portion, such as dielectric fill portion **206A**. Vertical interconnect structures **208A** and **208B** can be disposed in isolation structures **306A** and **306B**, respectively.

(56) Referring to FIG. 4, in some embodiments, semiconductor device **100** can have segmented isolation structures **306A** and **406**, which are spaced apart from each other along an X-axis. In some embodiments, isolation structure **406** can be similar to dielectric fill portion **106A**.

(57) Referring to FIG. 5, in some embodiments, isolation structure **506** can include a dielectric fill portion **506A** and a dielectric liner portion **506B**. Vertical interconnect structure **508** can be disposed in dielectric liner portion **506B**. Dielectric liner portion **506B** can be similar to dielectric liner portion **206A**, but unlike dielectric liner portion **206A**, dielectric liner portion **506B** can be in physical contact with S/D contact structures of both stacked FETs **102A** and **102B**. Vertical interconnect structure **508** can be similar to vertical interconnect structure **208A**, but unlike vertical interconnect structure **208A**, vertical interconnect structure **508** can be in physical contact with S/D

contact structures of both stacked FETs **102A** and **102B**. That is, vertical interconnect structure **508** can be electrically connected to S/D contact structures **122A2-1** and/or **122A2-2** and S/D contact structures **122B2-1** and/or **122B2-2**. Furthermore, unlike isolation structure **206**, the symmetry line of isolation structure **506** can be substantially aligned with standard cell boundary G-G.

(58) FIG. **6** is a flow diagram of an example method **600** for fabricating semiconductor devices **100** shown in FIGS. **1A-1H**, according to some embodiments. For illustrative purposes, the operations illustrated in FIG. **6** will be described with reference to the example fabrication process for fabricating semiconductor devices **100** as illustrated in FIGS. **7**, **8A-15B**, and **16A-22F**. FIG. **7** is an isometric view of semiconductor device **100** at a stage of fabrication, according to some embodiments. FIGS. **8A-15A** are cross-sectional views of semiconductor devices **100** along lines A-A of FIGS. **1A** and **1D-1H** at various stages of fabrication, according to some embodiments. FIGS. **8B-15B** are cross-sectional views of semiconductor devices **100** along lines H—H of FIGS. **1A** and **1D-1H** at various stages of fabrication, according to some embodiments.

(59) FIGS. **16A-22A** are cross-sectional views of semiconductor devices **100** along lines A-A of FIGS. **1A** and **1D-1H** at various stages of fabrication, according to some embodiments. FIGS. **16B-22B** are cross-sectional views of semiconductor devices **100** along lines B-B of FIGS. **1A** and **1D-1H** at various stages of fabrication, according to some embodiments. FIGS. **16C-22C** are cross-sectional views of semiconductor devices **100** along lines C-C of FIGS. **1A-1C** and **1H** at various stages of fabrication, according to some embodiments. FIGS. **16D-22D** are cross-sectional views of semiconductor devices **100** along lines D-D of FIGS. **1A-1C** and **1H** at various stages of fabrication, according to some embodiments. FIGS. **16E-22E** are cross-sectional views of semiconductor devices **100** along lines E-E of FIGS. **1A-1C** and **1H** at various stages of fabrication, according to some embodiments. FIGS. **16F-22F** are cross-sectional views of semiconductor devices **100** along lines F-F of FIGS. **1A-1C** and **1H** at various stages of fabrication, according to some embodiments.

(60) Operations can be performed in a different order or not performed depending on specific applications. It should be noted that method **600** may not produce a semiconductor device **100**. Accordingly, it is understood that additional processes can be provided before, during, and after method **600**, and that some other processes may only be briefly described herein. Elements in FIGS. **7-15B** and **16A-22F** with the same annotations as elements in FIGS. **1A-1H** are described above.

(61) Referring to FIG. **6**, in operation **605**, first and second stacks of superlattice structures are formed on first and second fin structures, respectively. For example, as shown in FIG. **7**, first and second stacks of superlattice structures **746A** and **746B** are formed on first and second fin structures **110A** and **110B**, respectively. First stack of superlattice structure **746A** can include first and second superlattice structures **746A1** and **746A2** and a nanostructured isolation layer **714A** disposed between first and second superlattice structures **746A1** and **746A2**. Second stack of superlattice structure **746B** can include first and second superlattice structures **746B1** and **746B2** and a nanostructured isolation layer **714B** disposed between first and second superlattice structures **746B1** and **746B2**.

(62) Each of superlattice structures **746A1-746A2** and **746B1-746B2** can include nanostructured layers **716** and **748**. In some embodiments, nanostructured layer **716** can include Si and nanostructured layer **748** can include SiGe. Nanostructured layers **716** can form nanostructured layers **116A1-116A2** and **116B1-116B** in subsequent processing. Nanostructured isolation layers **714A-714B** can form channel isolation layers **114A-114B** in subsequent processing. In some embodiments, the formation of first and second stacks of superlattice structures **746A** and **746B** can include epitaxially growing the materials of nanostructured layers **716** and **748** and depositing the material of nanostructured isolation layers **714A-714B** on substrate **104**, followed by a photolithography process and an etching process.

(63) Referring to FIG. **6**, in operation **610**, polysilicon structures are formed on the first and second

stacks of superlattice structures. For example, as shown in FIGS. **8A-8B**, polysilicon structures **818** are formed on first and second stacks of superlattice structures **746A** and **746B**.

(64) Referring to FIG. **6**, in operation **615**, bottom S/D regions are formed on the first and second fin structures and top S/D regions are formed on the bottom S/D regions. For example, as described with reference to FIGS. **9A-12B**, S/D regions **120A1-1**, **120A1-2**, **120B1-1**, and **120B1-2**, referred to as “bottom S/D regions,” are formed on fin structures **110A** and **110B**, and S/D regions **120A2-1**, **120A2-2**, **120B2-1**, and **120B2-2**, referred to as “top S/D regions,” are formed on the bottom S/D regions.

(65) The formation of the bottom S/D regions can include sequential operations of (i) forming S/D openings **952**, as shown in FIGS. **9A-9B**, by etching first and second stacks of superlattice structures **746A** and **746B** through openings **850** (shown in FIGS. **8A-8B**), and (ii) epitaxially growing a semiconductor layer with a first conductivity type dopants (e.g., p-type dopants) in S/D openings **952** to form the structures of FIGS. **10A-10B**. In some embodiments, inner spacers **128** can be formed between operations (i) and (ii) of the formation process of the bottom S/D regions, as shown in FIGS. **9A-9B**. The formation of the bottom S/D regions can be followed by the formation of ESLs **124A1-124B1** and ILD layers **126A1-126B1** on the bottom S/D regions, as shown in FIGS. **11A-11B**.

(66) The formation of the top S/D regions can include epitaxially growing a semiconductor layer with a second conductivity type dopants (e.g., n-type dopants) in S/D openings **952** to form the structures of FIGS. **12A-12B**. The semiconductor layer can epitaxially grow on the sidewalls of nanostructured layers **116A2** and **116B2** facing S/D openings **952**. During the epitaxial growth of the semiconductor layer, the semiconductor layer can extend laterally and vertically to substantially fill S/D openings **952**. The semiconductor layer does not grow epitaxially on the non-semiconductor material of inner spacers **128** and ILD layers **126A1-126B1**. As a result, in some embodiments, there can be air gaps (not shown) at the interfaces between the top S/D regions and inner spacers **128** and between the top S/D regions and ILD layers **126A1-126B1**. In some embodiments, top surfaces of the top S/D regions can be substantially coplanar with bottom surfaces of polysilicon structures **818**, as shown in FIGS. **12A-12B**. Following the formation of the top S/D regions, ESLs **124A2-124B2** and ILD layers **126A2-126B2** can be formed, as shown in FIGS. **13A-13B**.

(67) Referring to FIG. **6**, in operation **620**, bottom gate structures are formed on the first and second fin structures and top gate structures are formed on the bottom gate structures. For example, as described with reference to FIGS. **13A-15B** and **16A-16F**, gate structures **1518A-1**, **1518A-2**, and **1518A-3**, referred to as “bottom gate structures,” are formed on fin structures **110A** and **110B**, and gate structures **1518B-1**, **1518B-2**, and **1518B-3**, referred to as “top gate structures,” are formed on the bottom gate structures.

(68) The formation of the bottom gate structures can include sequential operations of (i) forming gate openings **1318**, as shown in FIGS. **13A-13B**, by etching polysilicon structures **818** and nanostructured layers **748** from the structures of FIGS. **12A-12B**, (ii) forming IL layers **132** on exposed regions of nanostructured layers **116A1-116A2** and **116B1-116B2** in gate openings **1318**, as shown in FIGS. **14A-14B**, (iii) depositing HK gate dielectric layers **134** on IL layers **132**, as shown in FIGS. **14A-14B**, (iv) depositing conductive layers **1436** on HK gate dielectric layers **134**, as shown in FIGS. **14A-14B**, and (v) etching conductive layers **1436** from gate openings **1318** above channel isolation layers **114A-114B** to form bottom gate structures as shown in FIGS. **15A-15B**. Conductive layers **1436** can have a material similar to that of conductive layers **136A1** discussed above. The formation of the top gate structures can include depositing conductive layers **1536** in gate openings **1318** above channel isolation layers **114A-114B**, followed by a chemical mechanical polishing (CMP) process to form the top gate structures shown in FIGS. **16A-16C** and **16E**. The formation of the top gate structures can be followed by the formation of ESLs **124A2-124B2** and ILD layers **126A2-126B2** on the top S/D regions, as shown in FIGS. **16D** and **16F**.

(69) Referring to FIG. 6, in operation 625, an isolation trench is formed to cut the top and bottom gate structures into shorter gate structures. For example, as described with reference to FIGS. 17A-18F, an isolation trench **1806** is formed through the top and bottom gate structures to cut the top and bottom gate structures into shorter gate structures of GAA FETs **102A1-102A2** and **102B1-102B2**. The formation of isolation trench **1806** through the top and bottom gate structures can be referred to as the “CMG process.”

(70) The formation of isolation trench **1806** can include sequential operations of (i) depositing a nitride layer **1754** on the structures of FIGS. 16A-16F, (ii) depositing a photoresist layer **1756** on nitride layer **1754**, (iii) forming openings **1757** in nitride layer **1754** and photoresist layer **1756** using a lithographic patterning process and an etching process, as shown in FIGS. 17A-17F, (iv) etching portions of the top and bottom gate structures, ESLs **124A1-124B2**, ILD layers **126A1-126B2**, and STI regions **112A-112B** through openings **1757** in patterned nitride layer **1754** and patterned photoresist layer **1756** to form isolation trench **1806**, as shown in FIGS. 18B-18F. Isolation trench **1806** is not visible in the cross-sectional view of FIG. 18A. In subsequent processing, (i) the portion of isolation trench **1806** visible in FIG. 18B can form the portions of isolation structure **106** and vertical interconnect structure **108** shown along line B-B in FIG. 1H, (ii) the portion of isolation trench **1806** visible in FIGS. 18C and 18D can form the portions of isolation structure **106** shown along lines C-C and D-D, respectively in FIG. 1H, and (iii) the portion of isolation trench **1806** visible in FIGS. 18E and 18F can form the portions of isolation structure **106** and vertical interconnect structure **108** shown along lines E-E and F—F, respectively in FIG. 1H.

(71) In some embodiments, the formation of isolation trench **1806** includes forming a first portion (shown in FIGS. 18C-18D) with a dimension **T1** and a second portion (shown in FIGS. 18E-18F) with a dimension **T2**, which is greater than dimension **T1**. The first portion of isolation trench **1806** corresponds to the subsequently-formed dielectric fill portion **106A** (shown in FIGS. 20C-20D) and the second portion of the isolation trench **1806** corresponds to the subsequently-formed dielectric liner portion **106B** (shown in FIGS. 20E-20F). In some embodiments, isolation trench **1806** can be formed with a height **H10** of about 180 nm to about 220 nm, dimension **T1** of about 10 nm to about 20 nm, and dimension **T2** of about 30 nm to about 40 nm. These ranges of height **H10** and dimension **T1-T2** can provide the first portion of isolation trench with an aspect ratio (e.g., **H10:T1**) of about 9 to about 22 and the second portion of isolation trench with an aspect ratio (e.g., **H10:T2**) of about 5 to about 7 without compromising device size and manufacturing cost. Within these aspect ratio ranges, fabrication defects (e.g., voids) in subsequently-formed isolation structure **106** and vertical interconnect structure **108** can be prevented or minimized.

(72) Referring to FIG. 6, in operation 630, an isolation structure with a fill portion and a liner portion is formed in the isolation trench. For example, as described with reference to FIGS. 19B-19F, isolation structure **106** with dielectric fill portion **106A** and dielectric liner portion **106B** are formed in the first and second portions of isolation trench **1806**, respectively. The formation of dielectric fill portion **106A** and dielectric liner portion **106B** can include depositing a nitride layer **1906** with a thickness **T3** of about 5 nm to about 10 nm on the structures of FIGS. 18A-18F to form the structures of FIGS. 19A-19F. This range of thickness **T3** can facilitate the merging of nitride layer **1906** deposited on sidewalls of the first portion of isolation trench **1806** to form dielectric fill portion **106A**, as shown in FIGS. 19C-19D. Below this range of thickness **T3**, nitride layer **1906** deposited on sidewalls of the first portion of isolation trench **1806** may not merge to fill the first portion of isolation trench **1806**. On the other hand, above this range of thickness **T3**, nitride layer **1906** may form an overhang at the opening of the first portion of isolation trench **1806** and constrict or close the opening of the first portion, thus preventing the first portion of isolation trench **1806** from being adequately filled with nitride layer **1906**.

(73) Referring to FIG. 6, in operation 635, a vertical interconnect structure is formed in the liner portion of the isolation structure. For example, as described with reference to FIGS. 20B and 20E-

20F, vertical interconnect structure **108** is formed in dielectric liner portion **106B**. The formation of vertical interconnect structure **108** can include depositing a conductive material on the structures of FIGS. **19A-19F**, followed by a CMP process to form the structures of FIGS. **20A-20F**. The deposition process fills the second portion of isolation trench **1806** with the conductive material, as shown in FIGS. **20B** and **20E-20F**. The CMP process can substantially coplanarize top surfaces of top gate structures, vertical interconnect structure **108**, dielectric fill portion **106A**, and dielectric liner portion **106B**. Vertical interconnect structure **108** is not visible in the cross-sectional views of FIGS. **20A** and **20C-20D**.

(74) Referring to FIG. **6**, in operation **640**, front-side S/D contact structures are formed on front-side surfaces of the top S/D regions. For example, as described with reference to FIGS. **21A**, **21D**, and **21F**, front-side S/D contact structures **122A2-1**, **122A2-2**, **122B2-1**, and **122B2-2** are formed on front-side surfaces of top S/D regions **120A2-1**, **120A2-2**, **120B2-1**, and **120B2-2**. Front-side S/D contact structures are not visible in the cross-sectional views of FIGS. **21B-21C** and **21E**. In some embodiments, gate capping layers **131A-131B** (shown in FIGS. **21A**, **21C**, and **21E**) can be formed prior to forming the front-side S/D contact structures.

(75) Referring to FIG. **6**, in operation **645**, back-side S/D contact structures are formed on back-side surfaces of the bottom S/D regions. For example, as described with reference to FIGS. **22A**, **22D**, and **22F**, back-side S/D contact structures **122A1-1**, **122A1-2**, **122B1-1**, and **122B1-2** are formed on back-side surfaces of bottom S/D regions **120A2-1**, **120A2-2**, **120B2-1**, and **120B2-2**. Back-side S/D contact structures are not visible in cross-sectional views of FIGS. **22B-22C** and **22E**. The formation of the back-side S/D contact structures can include sequential operations of (i) thinning or removing substrate **104**, as shown in FIGS. **22A-22F**, (ii) forming S/D contact openings (not shown) by removing portions of fin structures **110A-110B** under the bottom S/D regions, (iii) forming silicide layers **138** in the S/D contact openings, as shown in FIGS. **22A**, **22D**, and **22F**, and (iv) forming contact plugs **140** on silicide layers **138**, as shown in FIGS. **22A**, **22D**, and **22F**. In some embodiments, a portion of dielectric liner portion **106B** in physical contact with substrate **104**, as shown in FIGS. **21B-21F** can be removed during the thinning or removal of substrate **104**, as shown in FIGS. **22B-22F**.

(76) The present disclosure provides example semiconductor devices (e.g., semiconductor device **100**) with stacked FETs (e.g., stacked FETs **102A-102B**) and vertical interconnect structures (e.g., vertical interconnect structures **108**, **208A-208B**, and **508**) between adjacent stacked FETs. The present disclosure further provides example methods (e.g., method **600**) of forming the semiconductor devices. With the use of stacked FETs, the device density of ICs can be increased without aggressively scaling down the devices and compromising the electrical isolation between the devices in the IC. In some embodiments, each of the stacked FETs can include a stack of different conductivity type GAA FETs (e.g., GAA FETs **102A1-102A2**) and/or can include a stack of the same conductivity type GAA FETs. Each of the stacked FETs can further include a channel isolation layer (e.g., channel isolation layers **114A-114B**). The channel isolation layer can electrically isolate the channel regions of the stacked FETs from each other.

(77) In some embodiments, the vertical interconnect structures (also referred to as “conductive bridge structures” and “through-via structures”) can provide electrical connections between top and bottom GAA FETs (e.g., GAA FETs **102B1-102B2**) in a stacked FET (e.g., stacked FET **102B**). In some embodiments, the vertical interconnect structures can provide electrical connections between top GAA FETs and an interconnect structure on a back-side of the semiconductor device. In some embodiments, the vertical interconnect structures can provide electrical connections between bottom GAA FETs and an interconnect structure on a front-side of the semiconductor device. The vertical interconnect structures can be electrically connected to S/D contact structures (e.g., S/D contact structures **122B1-2** and **122B2-2**) of the top and/or bottom GAA FETs (e.g., GAA FETs **102B1-102B2**).

(78) The semiconductor device can further include isolation structures (e.g., isolation structures

106, 206, 306A-306B, 406, and 506) formed in a CMG process to “cut” long metal gate structures, extending over two or more of the stacked FETs, into shorter gate portions and to electrically isolate adjacent stacked FETs from each other. One or more of the isolation structures can include dielectric fill portions (e.g., dielectric fill portions **106A, 206A, 406, and 506A**) and dielectric liner portions (e.g., dielectric liner portions **106B, 206B1, 206B2, 306A-306B, and 506B**). The vertical interconnect structures can be formed in the dielectric liner portions to reduce the device area occupied by the isolation structures and the vertical interconnect structures in the semiconductor device compared to the device area occupied by vertical interconnect structures formed adjacent to isolation structures. The formation of the vertical interconnect structures in the isolation structures also relaxes the dimensional constraints on the vertical interconnect structures. Larger vertical interconnect structures can be formed in the isolation structures, which reduces the resistance of the vertical interconnect structures by about 60% to about 90% compared to the resistance of vertical interconnect structures formed adjacent to isolation structures.

(79) Furthermore, the vertical interconnect structures can be formed in the isolation structures with fewer process steps than the number of process steps involved in forming vertical interconnect structures adjacent to isolation structures. For example, the isolation trenches (e.g., isolation trench **1806**) formed using a photolithographic process and an etch process during the CMG process (e.g., operation **625** of method **600**) can be used to form both the isolation structures and the vertical interconnect structures. Portions of the isolation trenches can be filled with a dielectric material to form the dielectric fill portions (e.g., operation **630** of method **600**). Other portions of the isolation trenches can be lined with the dielectric material to form the dielectric liner portions (e.g., operation **630** of method **600**), which can be subsequently filled with a conductive material to form the vertical interconnect structures (e.g., operation **630** of method **600**). Thus, a single photolithographic process and a single etch process can be used to form both the vertical interconnect structures and the isolation structures, instead of the multiple photolithographic processes and multiple etch processes used for forming vertical interconnect structures adjacent to isolation structures.

(80) In some embodiments, a semiconductor device includes a first FET, a second FET, an isolation structure, and a conductive structure. The first FET includes a first fin structure, a first array of gate structures disposed on the first fin structure, and a first array of S/D regions disposed on the first fin structure. The second FET includes a second fin structure, a second array of gate structures disposed on the second fin structure, and a second array of S/D regions disposed on the second fin structure. The isolation structure includes a fill portion and a liner portion disposed between the first and second FETs and in physical contact with the first and second arrays of gate structures. The conductive structure is disposed in the liner portion of the isolation structure and conductively coupled to a S/D region of the second array of S/D regions.

(81) In some embodiments, a semiconductor device includes a first stacked FET, a second stacked FET, an isolation structure, and a vertical interconnect structure. The first stacked FET includes a first GAA FET of a first conductivity type and a second GAA FET of a second conductivity type disposed on the first GAA FET. The first and second conductivity types are different from each other. The second stacked FET includes a third GAA FET of the first conductivity type and a fourth GAA FET of the second conductivity type disposed on the third GAA FET. The isolation structure includes a dielectric fill portion and a dielectric liner portion disposed between the first and second stacked FETs. The vertical interconnect structure is surrounded by the dielectric liner portion and conductively coupled to S/D regions of the third and fourth GAA FETs.

(82) In some embodiments, a method includes forming first and second fin structures on a substrate, forming first and second arrays of S/D regions on the first and second fin structures, respectively, forming third and fourth arrays of S/D regions on the first and second arrays of S/D regions, respectively, forming an array of gate structures on the first and second fin structures, forming an isolation trench through the array of gate structures and between the first and second

arrays of S/D regions, depositing a nitride layer in the isolation trench to fill a first portion of the isolation trench and form a liner along sidewalls of a second portion of the isolation trench, and depositing a conductive material on the liner to fill the second portion of the isolation trench. (83) The foregoing disclosure outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device, comprising: a first field effect transistor (FET), comprising: a first fin structure, a first array of gate structures disposed on the first fin structure, and a first array of source/drain (S/D) regions; a second FET, comprising: a second fin structure, a second array of gate structures disposed on the second fin structure, and a second array of S/D regions; an isolation structure comprising a fill portion and a liner portion disposed between the first and second FETs and in physical contact with the first and second arrays of gate structures; and a conductive structure disposed in the liner portion of the isolation structure and conductively coupled to a S/D region of the second array of S/D regions.
2. The semiconductor device of claim 1, wherein the conductive structure is conductively coupled to first and second S/D regions of the second array of S/D regions, and wherein the second S/D region is disposed on the first S/D region.
3. The semiconductor device of claim 1, further comprising first and second contact structures disposed on first and second S/D regions of the second array of S/D regions, respectively, wherein the first and second contact structures are in physical contact with the conductive structure, and wherein the second S/D region is disposed on the first S/D region.
4. The semiconductor device of claim 1, wherein the first array of S/D regions is electrically isolated from the conductive structure.
5. The semiconductor device of claim 1, wherein sidewalls of the fill portion and liner portion facing the first array of gate structures are substantially coplanar with each other.
6. The semiconductor device of claim 1, wherein sidewalls of the fill portion and liner portion facing the second array of gate structures are non-coplanar with each other.
7. The semiconductor device of claim 1, wherein a portion of the liner portion and a portion of the conductive structure protrudes from a sidewall of the fill portion facing the second array of gate structures.
8. The semiconductor device of claim 1, wherein a width of the conductive structure is greater than a width of the S/D region of the second array of S/D regions or is greater than a gate pitch of the second array of gate structures.
9. The semiconductor device of claim 1, wherein the isolation structure further comprises a second liner portion coupled to the liner portion through the fill portion, and further comprising a second conductive structure disposed in the second liner portion.
10. The semiconductor device of claim 1, further comprising: a first contact structure disposed on a first S/D region of the first array of S/D regions; and a second contact structure disposed on a second S/D region of the second array of S/D regions, wherein the first and second contact structures are in physical contact with the conductive structure.
11. A semiconductor device, comprising: first and second fin structures; a first plurality of gate structures disposed on the first fin structure; a first plurality of source/drain regions disposed

adjacent to the first plurality of gate structures; a second plurality of gate structures disposed on the second fin structure; a second plurality of source/drain regions disposed adjacent to the second plurality of gate structures; a conductive structure coupled to the second plurality of source/drain regions and disposed between the first and second fin structures, and between the first and second plurality of gate structures, and between the first and second plurality of source/drain regions; and a dielectric structure, comprising: a fill portion disposed between the first and second plurality of source/drain regions; and a liner portion surrounding the conductive structure and disposed between the first and second plurality of gate structures, wherein the liner portion is in contact with the first and second plurality of gate structures and the conductive structure.

12. The semiconductor device of claim 11, wherein the fill portion is disposed adjacent to the conductive structure.

13. The semiconductor device of claim 11, further comprising a contact structure disposed on a surface of at least one of the first plurality of source/drain regions and in physical contact with a sidewall of the conductive structure.

14. The semiconductor device of claim 11, further comprising: a third source/drain region disposed on at least one of the first plurality of source/drain regions; a front contact structure disposed on a top surface of the third source/drain region and in physical contact with a sidewall of the conductive structure; and a back contact structure disposed on a bottom surface of at least one of the first plurality of source/drain regions and in physical contact with the sidewall of the conductive structure.

15. A semiconductor device, comprising: first and second fin structures; a first array of gate structures disposed on the first fin structure; a second array of gate structures disposed on the second fin structure; a first array of source/drain regions disposed on the first fin structure; a second array of source/drain regions disposed on the second fin structure; a conductive structure extending vertically between the first and second array of gate structures and between the first and second fin structures, wherein the conductive structure is conductively coupled to a source/drain region of the second array of source/drain regions; and an isolation structure, comprising: fill portion disposed between the first and second array of gate structures, and a liner portion surrounding the conductive structure and in contact with the first and second array of gate structures.

16. The semiconductor device of claim 15, wherein the fill portion comprises a nitride material.

17. The semiconductor device of claim 15, wherein the isolation structure further comprises another liner portion coupled to the liner portion through the fill portion.

18. The semiconductor device of claim 17, further comprising another conductive structure surrounded by the other liner portion.

19. The semiconductor device of claim 15, further comprising: a contact structure disposed on the source/drain region of the second array of source/drain regions and in physical contact with a sidewall of the conductive structure.

20. The semiconductor device of claim 15, further comprising: a contact structure disposed on the source/drain region of the second array of source/drain regions and in physical contact with a sidewall of the conductive structure and a sidewall of the fill portion.
